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Education

Addis Ababa University (AAU)

Bachelor of Science in Electrical and Computer Engineering

Addis Ababa, Ethiopia

Expected Graduation: May 2028

Projects

CodeDash | *React, Node.js, Express, MongoDB, Socket.io, Redis, Bull, JWT*

- Engineered a custom VS Code extension (VS Code API) that monitors local active files, keystrokes, and session times, utilizing a local offline queue to buffer data during network drops.
- Developed a scalable Node.js/Express backend that receives buffered session payloads and syncs active workspace data over WebSockets (Socket.io) to minimize API overhead.
- Integrated with the GitHub API to fetch and aggregate historical repository data, including commits, programming languages, and user activity metrics.
- Configured a secure GitHub Webhook receiver to handle real-time push and pull-request events, dynamically updating the database with live development logs.
- Implemented background task queues using Bull and Redis to handle asynchronous data synchronization without blocking the main event loop.
- Developed a responsive dark-mode React dashboard utilizing Recharts to visualize user productivity metrics, level progress, and language evolution.

Smart Logic Simulator | *Java, JavaFX, CSS, Object I/O Streams, SVGPath*

- Developed an interactive digital logic simulator in Java and JavaFX, separating UI rendering from underlying logic models.
- Engineered a custom real-time simulation engine using a high-frequency JavaFX AnimationTimer clock to propagate logic states across active gates.
- Designed custom binary serialization (*.ckt) using Object I/O streams, allowing users to save and reload circuit layouts locally.
- Programmed dynamic, vector-rendered connection wires using CubicCurve mathematical modeling to calculate and render wire routing and active logic states (glow effects) on the fly.
- Designed a responsive, dark-mode CSS stylesheet using custom variables, drop shadows, and neon transition properties to deliver a premium user interface experience.
- Rendered scalable, ANSI-standard logic gate vector shapes using inline JavaFX SVGPath strings.

Induction Motor Fault Detector | *Python, TensorFlow, scikit-learn, NumPy, Tkinter*

- Applied FFT to decompose motor current signals into the frequency domain and extract harmonic amplitudes at 50Hz, 150Hz, 250Hz, and 350Hz as diagnostic features for fault classification.
- Trained a Random Forest classifier (500 estimators) on 1,000 physics-informed synthetic samples achieving 100% accuracy on held-out test data; feature importance confirmed the 3rd harmonic as the primary fault indicator — consistent with real-world MCSA literature.
- Built a parallel 1D CNN (Conv1D → MaxPool × 2 → Dense) operating on raw 10,000-sample waveforms, achieving 100% validation accuracy on synthetic data; real-world performance would require validation against recorded motor signals.
- Developed a Tkinter desktop GUI for non-technical users to load CSV motor data, visualize time-domain signals and FFT spectra side-by-side, and receive confidence-scored diagnoses.

Technical Skills

Languages: JavaScript, Java, Python

Web Technologies: React, Node.js, Express, MongoDB, Socket.io, JWT, HTML5/CSS3, Tailwind CSS

Hardware & Tools: VS Code Extension API, Git, GitHub, Vercel, Netlify, Render, Bull, Redis, Logisim-Evolution

Familiar with: TensorFlow, scikit-learn, NumPy, Verilog, C, MATLAB